

Classification Metrics — Full Step-by-Step Solutions

Each solution follows: **understand the problem** → **identify TP/TN/FP/FN** → **formula** → **substitute** → **calculate** → **final answer** → **interpretation**.

Solution 1 — Fraud Detection

Step 1: Actual fraudulent transactions = 120. The model correctly detected 96.

Step 2: $FN = 120 - 96 = 24$.

Step 3: Actual legitimate = $2,000 - 120 = 1,880$. $FP = 56$, so $TN = 1,880 - 56 = 1,824$.

Confusion matrix values: $TP=96$, $FN=24$, $FP=56$, $TN=1,824$.

Step: Accuracy

Formula: $(TP + TN) / \text{Total}$

Substitute: $(96 + 1,824) / 2,000$

Calculate: $1,920 / 2,000$

Final Answer: **96.00%**

What it means: The percentage of all transactions classified correctly.

Step: Precision

Formula: $TP / (TP + FP)$

Substitute: $96 / (96 + 56)$

Calculate: $96 / 152$

Final Answer: **63.16%**

What it means: Of transactions predicted as fraud, the percentage that were actually fraud.

Step: Recall / Sensitivity

Formula: $TP / (TP + FN)$

Substitute: $96 / (96 + 24)$

Calculate: $96 / 120$

Final Answer: **80.00%**

What it means: Of actual fraud cases, the percentage detected.

Step: Specificity

Formula: $TN / (TN + FP)$

Substitute: $1,824 / (1,824 + 56)$

Calculate: $1,824 / 1,880$

Final Answer: **97.02%**

What it means: Of legitimate transactions, the percentage correctly identified.

Step: FPR

Formula: $FP / (FP + TN)$

Substitute: $56 / (56 + 1,824)$

Calculate: $56 / 1,880$

Final Answer: 2.98%

What it means: The percentage of legitimate transactions incorrectly flagged as fraud.

Step: FNR

Formula: $FN / (FN + TP)$

Substitute: $24 / (24 + 96)$

Calculate: $24 / 120$

Final Answer: 20.00%

What it means: The percentage of actual fraud cases missed.

Step: F1 Score

Formula: $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Substitute: $2 \times (0.6316 \times 0.8000) / (0.6316 + 0.8000)$

Calculate: 0.7059

Final Answer: 70.59%

What it means: Balances Precision and Recall.

Decision: Since the bank wants to minimize missed fraud, prioritize Recall.

Solution 2 — Medical Screening

Step 1: Actual positives = 800. TP = 720, therefore FN = $800 - 720 = 80$.

Step 2: Total predicted positives = 900. Therefore FP = $900 - 720 = 180$.

Step 3: Actual negatives = $10,000 - 800 = 9,200$. TN = $9,200 - 180 = 9,020$.

Step: Precision

Formula: $TP / (TP + FP)$

Substitute: $720 / (720 + 180)$

Calculate: $720 / 900$

Final Answer: **80.00%**

What it means: How reliable positive predictions are.

Step: Recall

Formula: $TP / (TP + FN)$

Substitute: $720 / (720 + 80)$

Calculate: $720 / 800$

Final Answer: **90.00%**

What it means: How many actual disease cases are detected.

Step: Specificity

Formula: $TN / (TN + FP)$

Substitute: $9,020 / (9,020 + 180)$

Calculate: $9,020 / 9,200$

Final Answer: **98.04%**

What it means: How many actual healthy cases are correctly identified.

Step: F1 Score

Formula: $2PR / (P + R)$

Substitute: $2 \times (0.8000 \times 0.9000) / (0.8000 + 0.9000)$

Calculate: 0.8471

Final Answer: **84.71%**

What it means: Balances Precision and Recall.

Decision: Recall is most important when failing to identify a diseased patient is the main concern.

Solution 3 — Two Classifiers

Model A: TP=360, TN=4,460, FP=140, FN=40.

Step: Model A Precision

Formula: $TP/(TP+FP)$

Substitute: $360/(360+140)$

Calculate: $360/500$

Final Answer: 72.00%

What it means: Reliability of positive predictions.

Step: Model A Recall

Formula: $TP/(TP+FN)$

Substitute: $360/(360+40)$

Calculate: $360/400$

Final Answer: 90.00%

What it means: Defective products detected.

Step: Model A F1

Formula: $2PR/(P+R)$

Substitute: $2 \times (0.72 \times 0.90) / (0.72 + 0.90)$

Calculate: 0.8000

Final Answer: 80.00%

What it means: Balance of Precision and Recall.

Model B: TP=380, TN=4,300, FP=300, FN=20.

Step: Model B Precision

Formula: $TP/(TP+FP)$

Substitute: $380/(380+300)$

Calculate: $380/680$

Final Answer: 55.88%

What it means: Reliability of positive predictions.

Step: Model B Recall

Formula: $TP/(TP+FN)$

Substitute: $380/(380+20)$

Calculate: $380/400$

Final Answer: 95.00%

What it means: Defective products detected.

Step: Model B F1

Formula: $2PR/(P+R)$

Substitute: $2 \times (0.5588 \times 0.95) / (0.5588 + 0.95)$

Calculate: 0.7059

Final Answer: 70.59%

What it means: Balance of Precision and Recall.

Decision: Choose Model B if missing a defective product is more costly because Recall is higher. Choose Model A if false alarms are more costly because Precision is higher.

Solution 4 — Imbalanced Cybersecurity

Step 1: Actual attacks = 200. TP=150, so FN = 200–150 = **50**.

Step 2: Actual normal events = 20,000–200 = 19,800. FP=300, so TN = 19,800–300 = **19,500**.

Step: Accuracy

Formula: $(TP+TN)/Total$

Substitute: $(150+19,500)/20,000$

Calculate: $19,650/20,000$

Final Answer: **98.25%**

What it means: Overall correctness.

Step: Precision

Formula: $TP/(TP+FP)$

Substitute: $150/(150+300)$

Calculate: $150/450$

Final Answer: **33.33%**

What it means: Reliability of attack alerts.

Step: Recall

Formula: $TP/(TP+FN)$

Substitute: $150/(150+50)$

Calculate: $150/200$

Final Answer: **75.00%**

What it means: Actual attacks detected.

Step: Specificity

Formula: $TN/(TN+FP)$

Substitute: $19,500/(19,500+300)$

Calculate: $19,500/19,800$

Final Answer: **98.48%**

What it means: Normal events correctly identified.

Step: F1 Score

Formula: $2PR/(P+R)$

Substitute: $2 \times (0.3333 \times 0.7500) / (0.3333 + 0.7500)$

Calculate: 0.4615

Final Answer: **46.15%**

What it means: Balance of Precision and Recall.

Interpretation: Accuracy looks very high because normal events dominate. Precision and F1 show that attack detection is much weaker.

Solution 5 — Reverse Calculation: Precision and Recall

Step 1: Find TP and FN.

Formula: $\text{Recall} = \text{TP}/(\text{TP}+\text{FN})$.

$\text{TP} = \text{Recall} \times \text{Actual Positives} = 0.75 \times 200 = \mathbf{150}$.

$\text{FN} = 200 - 150 = \mathbf{50}$.

Step 2: Find FP.

Formula: $\text{Precision} = \text{TP}/(\text{TP}+\text{FP})$.

$0.60 = 150/(150+\text{FP})$. Therefore $150+\text{FP} = 150/0.60 = 250$.

$\text{FP} = 250 - 150 = \mathbf{100}$.

Step 3: Find TN. Actual negatives = $1,000-200=800$. $\text{TN} = 800-100 = \mathbf{700}$.

Step: Specificity

Formula: $\text{TN}/(\text{TN}+\text{FP})$

Substitute: $700/(700+100)$

Calculate: $700/800$

Final Answer: **87.50%**

What it means: Actual negatives correctly identified.

Step: F1 Score

Formula: $2\text{PR}/(\text{P}+\text{R})$

Substitute: $2 \times (0.60 \times 0.75)/(0.60+0.75)$

Calculate: 0.6667

Final Answer: **66.67%**

What it means: Balance of Precision and Recall.

Complete confusion matrix: $\text{TP}=150$, $\text{FN}=50$, $\text{FP}=100$, $\text{TN}=700$.

Solution 6 — Reverse Calculation: Accuracy

Step 1: $FN = \text{Actual positives} - TP = 500 - 400 = 100$.

Formula: $\text{Correct predictions} = \text{Accuracy} \times \text{Total}$.

$\text{Correct} = 0.90 \times 2,000 = 1,800$.

Step 2: $TN = \text{Correct} - TP = 1,800 - 400 = 1,400$.

Step 3: $\text{Actual negatives} = 2,000 - 500 = 1,500$. $FP = 1,500 - 1,400 = 100$.

Step: Precision

Formula: $TP/(TP+FP)$

Substitute: $400/(400+100)$

Calculate: $400/500$

Final Answer: 80.00%

What it means: Reliability of positive predictions.

Step: Recall

Formula: $TP/(TP+FN)$

Substitute: $400/(400+100)$

Calculate: $400/500$

Final Answer: 80.00%

What it means: Actual positives detected.

Step: Specificity

Formula: $TN/(TN+FP)$

Substitute: $1,400/(1,400+100)$

Calculate: $1,400/1,500$

Final Answer: 93.33%

What it means: Actual negatives correctly identified.

Step: F1 Score

Formula: $2PR/(P+R)$

Substitute: $2 \times (0.80 \times 0.80) / (0.80 + 0.80)$

Calculate: 0.8000

Final Answer: 80.00%

What it means: Balance of Precision and Recall.

Complete confusion matrix: $TP=400$, $FN=100$, $FP=100$, $TN=1,400$.

Solution 7 — Threshold Comparison

T1: TP=160, FN=40, FP=80, TN=720.

Step: T1 Precision

Formula: $TP/(TP+FP)$

Substitute: $160/(160+80)$

Calculate: $160/240$

Final Answer: 66.67%

What it means: Reliability of positive predictions.

Step: T1 Recall

Formula: $TP/(TP+FN)$

Substitute: $160/(160+40)$

Calculate: $160/200$

Final Answer: 80.00%

What it means: Defaulters detected.

Step: T1 Specificity

Formula: $TN/(TN+FP)$

Substitute: $720/(720+80)$

Calculate: $720/800$

Final Answer: 90.00%

What it means: Non-defaulters correctly identified.

Step: T1 F1

Formula: $2PR/(P+R)$

Substitute: $2 \times (0.6667 \times 0.80) / (0.6667 + 0.80)$

Calculate: 0.7273

Final Answer: 72.73%

What it means: Balance of Precision and Recall.

T2: TP=185, FN=15, FP=160, TN=640.

Step: T2 Precision

Formula: $TP/(TP+FP)$

Substitute: $185/(185+160)$

Calculate: $185/345$

Final Answer: 53.62%

What it means: Reliability of positive predictions.

Step: T2 Recall

Formula: $TP/(TP+FN)$

Substitute: $185/(185+15)$

Calculate: $185/200$

Final Answer: 92.50%

What it means: Defaulters detected.

Step: T2 Specificity

Formula: $TN/(TN+FP)$

Substitute: $640/(640+160)$

Calculate: $640/800$

Final Answer: 80.00%

What it means: Non-defaulters correctly identified.

Step: T2 F1

Formula: $2PR/(P+R)$

Substitute: $2 \times (0.5362 \times 0.925) / (0.5362 + 0.925)$

Calculate: 0.678?

Final Answer: 67.80%

What it means: Balance of Precision and Recall.

Decision: T2 detects more defaulters because Recall is higher. T1 is preferable when false alarms/interventions are more costly because Precision and Specificity are higher.

Solution 8 — Quality Control

Step 1: $TP=400$. $FP = 650-400 = 250$.

Step 2: $FN = 500-400 = 100$.

Step 3: Actual non-defective = $50,000-500 = 49,500$. $TN = 49,500-250 = 49,250$.

Step: Precision

Formula: $TP/(TP+FP)$

Substitute: $400/(400+250)$

Calculate: $400/650$

Final Answer: **61.54%**

What it means: Reliability of defective-product alerts.

Step: Recall

Formula: $TP/(TP+FN)$

Substitute: $400/(400+100)$

Calculate: $400/500$

Final Answer: **80.00%**

What it means: Defects detected.

Step: Specificity

Formula: $TN/(TN+FP)$

Substitute: $49,250/(49,250+250)$

Calculate: $49,250/49,500$

Final Answer: **99.49%**

What it means: Good products correctly cleared.

Step: Accuracy

Formula: $(TP+TN)/Total$

Substitute: $(400+49,250)/50,000$

Calculate: $49,650/50,000$

Final Answer: **99.30%**

What it means: Overall correctness.

Step: F1 Score

Formula: $2PR/(P+R)$

Substitute: $2 \times (0.6154 \times 0.80) / (0.6154 + 0.80)$

Calculate: 0.6957

Final Answer: **69.57%**

What it means: Balance of Precision and Recall.

Decision: If defective products reaching customers are unacceptable, emphasize Recall.

Solution 9 — FPR/FNR Comparison

Actual phishing = 2,000; actual legitimate = 8,000.

Model X: Recall=92%, Specificity=97%.

TP = $0.92 \times 2,000 = 1,840$; FN = $2,000 - 1,840 = 160$.

TN = $0.97 \times 8,000 = 7,760$; FP = $8,000 - 7,760 = 240$.

Step: Model X FPR

Formula: $1 - \text{Specificity}$

Substitute: $1 - 0.97$

Calculate: 0.03

Final Answer: 3.00%

What it means: Legitimate emails incorrectly blocked.

Step: Model X FNR

Formula: $1 - \text{Recall}$

Substitute: $1 - 0.92$

Calculate: 0.08

Final Answer: 8.00%

What it means: Phishing emails missed.

Step: Model X Precision

Formula: $\text{TP} / (\text{TP} + \text{FP})$

Substitute: $1,840 / (1,840 + 240)$

Calculate: $1,840 / 2,080$

Final Answer: 88.46%

What it means: Reliability of phishing alerts.

Step: Model X F1

Formula: $2\text{PR} / (\text{P} + \text{R})$

Substitute: $2 \times (0.8846 \times 0.92) / (0.8846 + 0.92)$

Calculate: 0.9018

Final Answer: 90.18%

What it means: Balance of Precision and Recall.

Model Y: Recall=97%, Specificity=92%.

TP = $0.97 \times 2,000 = 1,940$; FN = 60.

TN = $0.92 \times 8,000 = 7,360$; FP = 640.

Step: Model Y FPR

Formula: $1 - \text{Specificity}$

Substitute: $1 - 0.92$

Calculate: 0.08

Final Answer: 8.00%

What it means: Legitimate emails incorrectly blocked.

Step: Model Y FNR

Formula: $1 - \text{Recall}$

Substitute: $1 - 0.97$

Calculate: 0.03

Final Answer: 3.00%

What it means: Phishing emails missed.

Step: Model Y Precision

Formula: $\text{TP}/(\text{TP} + \text{FP})$

Substitute: $1,940/(1,940 + 640)$

Calculate: $1,940/2,580$

Final Answer: 75.19%

What it means: Reliability of phishing alerts.

Step: Model Y F1

Formula: $2\text{PR}/(\text{P} + \text{R})$

Substitute: $2 \times (0.7519 \times 0.97)/(0.7519 + 0.97)$

Calculate: 0.8468

Final Answer: 84.68%

What it means: Balance of Precision and Recall.

Decision: Model Y is preferred under the stated cost condition because it reduces the more expensive error: missed phishing (FNR 3% vs 8%).

Solution 10 — Mixed Missing Values

Step 1: Actual positives = 1,000; actual negatives = 4,000–1,000 = **3,000**.

Formula: Specificity = $TN/(TN+FP)$

$TN = 0.80 \times 3,000 = \mathbf{2,400}$. Therefore $FP = 3,000 - 2,400 = \mathbf{600}$.

Formula: Precision = $TP/(TP+FP)$

$0.625 = TP/(TP+600)$. Solving: $0.625(TP+600)=TP \rightarrow 0.625TP+375=TP \rightarrow 0.375TP=375 \rightarrow TP=\mathbf{1,000}$.

$FN = \text{Actual positives} - TP = 1,000 - 1,000 = \mathbf{0}$.

Step: Recall

Formula: $TP/(TP+FN)$

Substitute: $1,000/(1,000+0)$

Calculate: $1,000/1,000$

Final Answer: 100.00%

What it means: Actual positives detected.

Step: FNR

Formula: $FN/(FN+TP)$

Substitute: $0/(0+1,000)$

Calculate: $0/1,000$

Final Answer: 0.00%

What it means: Actual positives missed.

Step: Accuracy

Formula: $(TP+TN)/\text{Total}$

Substitute: $(1,000+2,400)/4,000$

Calculate: $3,400/4,000$

Final Answer: 85.00%

What it means: Overall correctness.

Step: F1 Score

Formula: $2PR/(P+R)$

Substitute: $2 \times (0.625 \times 1)/(0.625+1)$

Calculate: 0.7692

Final Answer: 76.92%

What it means: Balance of Precision and Recall.

Complete confusion matrix: $TP=1,000$, $FN=0$, $FP=600$, $TN=2,400$.